Author Response 2

Responses to the reviewers

Reviewer: 3

Comments to the Author

The authors responded adequately to my questions except one. There is a difference between neutralizing antibodies and total IgG. As far as I understand, the trials that are cited regarding antibodies, assessed total IgG by a ELISA test, i.e. without doing a neutralization test on COVID cultures. The authors should check this point in each trial and if it remains true that no trial assessed the neutralization capacity of the plasma, they should state it in the result and discuss it in the discussion sections. This is one of the reasons to promote trials of better quality, i.e. using a better calibrated product regarding its active component.

R: Thank you for pointing on this issue. In this new version we have addressed it in the results section in the text and in Table 1, in the discussion, and finally in the conclusion, as follows:

In the Results:

All studies quantified anti-covid-19 antibody titers, but only Li 19 and Agarwal 18 reported tests for quantifying neutralizing antibodies using two cell culture methodologies. Also, Li 19 reported the correlation of serum neutralization titer of 1:80 is approximately equivalent to a titer of 1: 1280 for S-RBD-specific IgG.

In table 1:

Table 1. Main characteristics of the included studies

Author	Study site	Design	Sample	Intervention	Neutralizing antibodies test	Control	Antiviral treatment	Primary outcomes	Secondary outcomes
Li [19]	China	Randomized, open-label, multicentre, clinical trial	52 intervention, 51 control	Convalescent plasma 4 to 13 mL/kg of recipient body weight. 96% received a single dose transfusion. IgG titer greater than 1:640	TCID50 (Tissue Culture Median Infectious Dose) Reed-Muench to calculate neutralizing anti-group titers A serum neutralization titer of 1:80 is roughly equivalent to a titer of 1: 1280 for S-RBD-specific IgG	Standard care: symptomatic control	Yes, both groups	Clinical improvement at day 7, 14 and 28	mortality at day 28, hospital stay, virological clearance,
Agarwal [18]	India	Randomized, open-label, multicentre, clinical trial	235 intervention, 229 control	2 doses of 200 mL convalescent plasma, transfused 24 hours apart. IgG titer greater than 1:1280	Microneutralization Assays in Cell Culture (TCID50)	Standard care following institutional protocol	Yes, only the control group	Progression to severe disease and mortality at day 28	Clinical improvement at day 7, O2 requirement, respiratory support, ventilation requirement, organ failure, virological clearance, levels of biomarkers, and vasopressor support
Abolghasemi [20]	Iran	Non-randomized, multicentre, clinical trial	115 intervention, 74 control	1 dose of 500 ml of convalescent plasma. The second dose of 500 ml was administered if the patient did not show any improvement after 24 hours. Titer cut off index higher than 1:1	No tests	Standard care	Yes, both groups	Mortality and hospital stay	Ventilation requirement and adverse events

In the Discussion:

The use of neutralizing antibody tests19 is strongly recommended to ensure the quality of the plasma transfused to patients, without forgetting that the host's response depends on particular phenomena, such as Original Antigenic Sin (OAS). This phenomenon shows that previous exposure to a similar virus triggers a strong but not specifically neutralizing immune response, so having high antibody titers or a strong immune response does not indicate that the virus is being neutralized.

In the Conclusion

The current evidence suggests that transfusion with convalescent plasma cannot reduce the mortality or ventilation requirement in hospitalized patients diagnosed with SARS-CoV-2 infection. More controlled clinical trials conducted with methodologies that ensure a low risk of bias, **with neutralizing antibody tests to ensure a good quality of plasma is still needed.**